

beacon. From these observations, time of the closest approach to the distress beacon and hence the location of the distress beacon are computed. GOES-E, GOES-W and INSAT satellites in geostationary earth orbit facilitate determination of the location of distress beacon with the help of GPS receiver integrated with the beacon. INSAT contributes for this system over the Indian Ocean region.

Satellite personal communication

In the 1990s, the idea of satellite personal communication entered the field of satellite communication. The progression of INTELSAT through seven generations clearly demonstrated reduction in the size of ground terminals inversely to increase in the size of satellites. The ultimate in this development would be a receive terminal that could be handheld. As an alternative to increasing the size and power of the GEO satellite, the satellite could be brought down to medium- or low-earth orbit to facilitate communication with a small terminal on ground.

After much debate and several studies, a few innovative systems were established. Iridium's satellite network is the only truly global communications network established with a constellation of 66 LEO satellites and has coverage available including the Polar regions. The system provides voice, paging, 160-character two-way short-text messaging (SMS), emergency 911 service (dial sequence 00-911) and Internet access services to subscribers anywhere on the surface of the earth.

Globalstar, deployed with a constellation of 48 LEO satellites, has over 250,000 subscribers in 120 countries. Due to problems with the Globalstar satellite network announced early in 2007, Globalstar is at present suited only for land-based users in the lower 48 US states and southern Canada. Coverage exists in most of the Americas and their coastal waters. One can dial a convenient US mobile number without expensive international long-



Iridium satellite

distance. In addition to voice services, 9600bps data service is available for e-mail and Internet access. With required special hardware and software, even faxes can be sent and received.

Thuraya system uses geosynchronous satellites and handheld satellite telephones to provide voice, Internet/data, fax, SMS and GPRS communications. Remote location-determination services via (GPS) are also available. Satellite coverage is now available in hundreds of countries spanning a large region of northern, central and western Africa, the Middle East, Central Asia and the Indian subcontinent. Certain models of the handheld phones also have terrestrial GSM cellular capability.

Satellite mobile systems are not preferred where terrestrial mobile systems have good coverage.

VSAT communication

VSAT stands for 'very small aperture terminal system.' The earth station antenna is typically less than 2.4m in diameter. The small receive-only terminal of direct broadcast satellite or DTH system could be termed 'VSAT,' but when we say VSAT, we mean private networks, mostly providing two-way communication facility. Typical user

groups include banking and financial institutions, ATMs, airlines, hotel booking agencies and large retail stores with geographically dispersed outlets.

Basically, a VSAT network consists of a hub station that provides broadcast facility to all the VSATs in the network. The VSATs themselves access the satellite in some form of multiple-access mode. The hub station may be operated by a service provider and shared by a number of users. Each user organisation has exclusive access to its own VSAT network.

Time-division multiplexing is the normal mode of transmission from the hub to VSATs. The access from VSATs to hub is more complex and various methods are used, which are proprietary. The most popular access method is FDMA. TDMA can also be used; since the data transfer from VSATs is of busy nature, a form of demand-assigned multiple access (DAMA) is employed in some systems.

VSAT systems operate in star configuration, which means the connection of one VSAT with another VSAT is through the hub involving two satellite hops and consequent delay. However, today mesh connections are available where VSATs connect with one an-